Partha Chakraborty

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SUMMARY

Full-stack engineer with 2+ years at startups, building web and mobile products in React, Next.js, and TypeScript on Python, FastAPI, and Node.js backends. AI work includes LoRA fine-tuning of Qwen2.5-7B and a deployed multi-agent codebase RAG application. Completing an M.S. in AI and Machine Learning at the University at Buffalo, expected January 2027.

EXPERIENCE

TurboMail **Remote**
AI Engineer Intern *May 2026 – Aug 2026*

- Owned production features across the stack in React, TypeScript, Python, FastAPI, SQLAlchemy, and SQL.
- Designed the database schema and APIs for subscription and payment tracking, then built the customer-facing React interface.
- Wrote bulk email deletion and folder-move APIs, plus email-body sanitization and thread scoping across multiple accounts.
- Reduced manual unsubscribe work with FastAPI background tasks that parse RFC 8058 List-Unsubscribe headers.
- Fine-tuned Qwen2.5-7B with LoRA and PEFT in PyTorch on a custom email dataset for an in-product feature.

Bird Vision AB **Remote**
Software Developer *Jun 2023 – Jul 2025*

- Ranked second in team code contributions within a year while shipping frontend and backend features across three products.
- Redesigned Eboka's responsive customer interface with the team using React, Next.js, Redux, and Tailwind CSS.
- Built and deployed a cross-platform employee app with React Native and Expo.
- Wrote Cypress tests covering key user flows in the web app.
- Integrated third-party APIs with Node.js and Express.js, taking backend features through production deployment.

EDUCATION

University at Buffalo **New York, USA**
M.S. in Artificial Intelligence and Machine Learning *GPA 3.9/4.0 | Expected Jan 2027*
Coursework: Algorithms, Database Systems, Data-Intensive Computing, Machine Learning

Presidency University **Bengaluru, India**
B.Tech. in Computer Science and Engineering *GPA 8.14/10 | 2018 – 2022*
Coursework: Computer Networks

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, HTML, CSS
Frontend: React, Next.js, Redux, Tailwind CSS, responsive design, React Native, Expo, Cypress
Backend: FastAPI, Node.js, Express.js, REST APIs, HTTP, SQLAlchemy, background tasks, third-party API integrations
AI and ML: PyTorch, LoRA, PEFT, multi-agent RAG, Gemini API, embeddings, vector search, BM25, cross-encoder reranking, tool calling, NumPy, Pandas, scikit-learn, NetworkX
Data and tools: PostgreSQL, MySQL, MongoDB, Firebase, ChromaDB, Qdrant, Tree-sitter, Docker, Linux, Git, Streamlit, Altair
AI coding tools: Cursor, Claude Code, GitHub Copilot

PROJECTS

AI Codebase Intelligence [Live Demo] [GitHub] Python, Gemini, ChromaDB, Streamlit

- Deployed a multi-agent RAG app for public GitHub and local repositories with four specialized agents: Architect, Code Reviewer, Bug Finder, and Documentation.
- Implemented Tree-sitter chunking across six languages, hybrid semantic/BM25/symbol retrieval, and cross-encoder reranking to ground answers in source code.
- Built a source explorer with file paths, line ranges, and citation checks; achieved 87.5% Hit@5 and 0.750 MRR on an initial 8-question retrieval evaluation.

GraphBench [Live Demo] Python, NetworkX, LadybugDB, Streamlit, Altair

- Built a dependency analysis tool using PageRank, betweenness centrality, cycle detection, and K-core analysis to identify critical dependencies, bottlenecks, and cycles.
- Validates CSV data before loading; an interactive explorer and JSON/CSV reports explain findings in plain language.
- Simulates failures by tracing direct and indirect dependents, blast radius, impact depth, and shortest dependency paths.

Isolation Forest [Live Demo] Python, FastAPI, Next.js, NumPy

- Implemented the algorithm from its original paper: recursive tree construction, training, and anomaly scoring.
- Built a FastAPI backend and Next.js dashboard for uploading data, running the model, and exploring anomalies.

CERTIFICATIONS

Hugging Face Agents Course — Unit 1: Fundamentals of Agents [Certificate] Oct 2026